ArchiveMind AI: Intelligent Knowledge Graph and Semantic Search Platform for Government Documents

Problem Statement

Government departments generate thousands of documents such as policies, schemes, circulars, reports, acts, notifications, and budget documents. These documents are usually stored as PDFs, making it difficult for citizens and government officials to quickly find relevant information.

Traditional keyword-based search systems often fail to understand the meaning and relationships between documents, resulting in inefficient information retrieval.

Proposed Solution

ArchiveMind AI is an AI-powered platform that transforms government documents into an intelligent knowledge repository. The system automatically extracts information from uploaded documents, identifies entities and relationships, builds a knowledge graph, and enables semantic search and natural language question answering.

Instead of manually reading hundreds of pages, users can simply ask questions and receive accurate answers with document references.

Objectives

- Digitize and organize government documents.
- Extract important entities such as ministries, schemes, organizations, locations, budgets, and dates.
- Identify relationships between entities.
- Build a knowledge graph for connected information discovery.
- Implement semantic search to understand user intent.
- Provide an AI-powered chatbot for natural language queries.
- Improve accessibility and usability of government information.

System Workflow

Step 1: Document Upload

Government officers upload PDF documents into the system.

Step 2: Text Extraction

The system extracts textual content from uploaded documents.

Step 3: Entity Recognition

AI identifies important entities such as:

- Ministries
- Government Schemes
- Organizations
- Dates
- Budgets
- Locations

Step 4: Relationship Extraction

Relationships between entities are identified.

Example:

Ministry of MSME → launched → PM Vishwakarma Scheme

Step 5: Knowledge Graph Creation

Extracted entities and relationships are stored in a graph database to create an interconnected knowledge network.

Step 6: Semantic Search

User queries are converted into embeddings and matched based on meaning rather than exact keywords.

Step 7: AI Question Answering

Users can ask questions in natural language and receive accurate answers generated from relevant documents.

Example Scenario

Uploaded Document:

"The Ministry of MSME launched the PM Vishwakarma Scheme in 2023 with a budget allocation of ₹13,000 crore."

Knowledge Graph:

MSME → launched → PM Vishwakarma Scheme

PM Vishwakarma Scheme → budget → ₹13,000 crore

User Query:

"Which ministry launched PM Vishwakarma?"

System Response:

"The Ministry of MSME launched the PM Vishwakarma Scheme in 2023."

Key Features

- Government document repository
- Automated document processing
- Knowledge graph generation
- Semantic search
- AI-powered chatbot
- Document summarization
- Relationship visualization
- Source-based answer generation
- Multi-document information retrieval

Technology Stack

Frontend

- React.js

Backend

- Spring Boot

Database

- PostgreSQL

Knowledge Graph Database

- Neo4j

NLP and AI

- Python

- SpaCy
- Sentence Transformers

Vector Search

- FAISS

AI Model

- Llama 3 / Open-source LLM

Expected Outcomes

- Faster access to government information.
- Reduced manual effort in document analysis.
- Improved decision-making for government officials.
- Enhanced transparency for citizens.
- Intelligent discovery of relationships across documents.

Innovation

ArchiveMind AI combines Knowledge Graphs, Semantic Search, Natural Language Processing, Vector Databases, and Retrieval-Augmented Generation (RAG) to create an intelligent government document discovery platform. The system enables users to interact with government documents through a conversational interface instead of traditional manual search methods.

One-Line Project Summary

ArchiveMind AI is an AI-powered government document intelligence platform that converts uploaded documents into a searchable knowledge graph and provides semantic search and chatbot-based question answering using RAG technology.